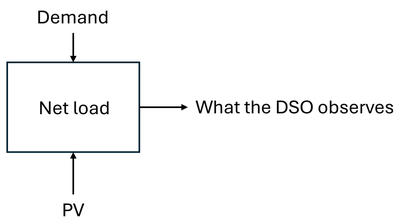
PRISM: Photovoltaic Recognition through Intelligent Similarity Mechanisms

Thomas Hull | Dr Xueqin Liu | Dr Robert Best

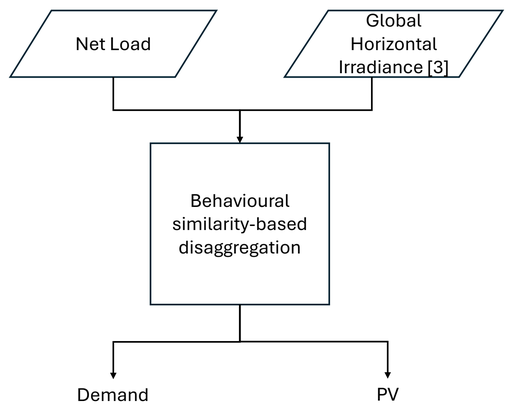
The Observability Challenge

Increasing adoption of behind-the-meter technologies is reducing visibility of underlying demand in low voltage networks [1]. Monitoring typically observes aggregate net load rather than demand and embedded generation independently, making it difficult to determine what is happening behind the meter [2].



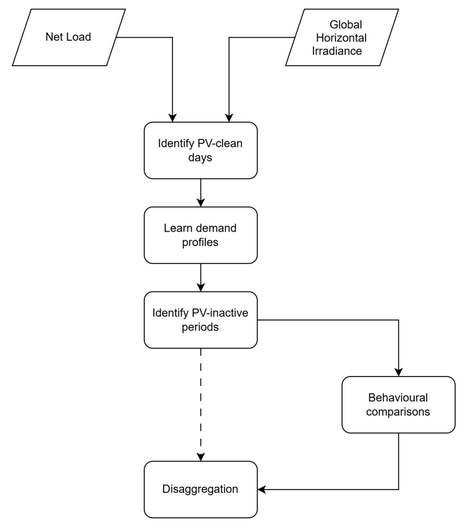
Project Aim

This project investigates whether behavioural similarity can provide additional information for disaggregating hidden behind-the-meter technologies from aggregate low-voltage measurements.

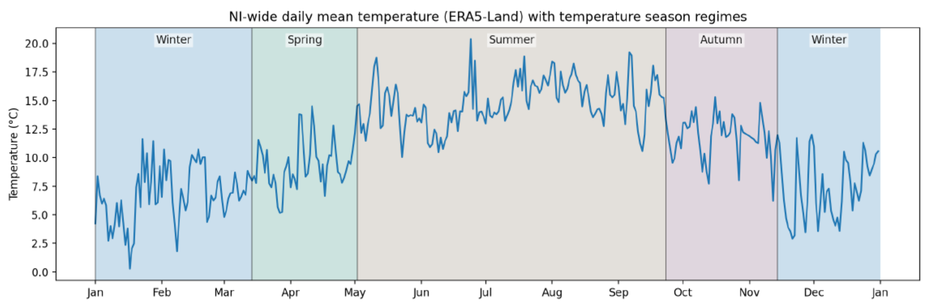


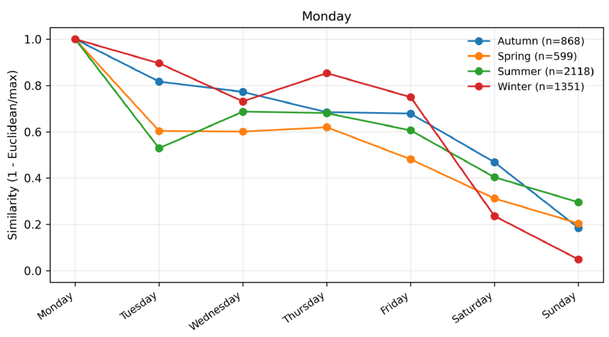
No additional customer-level monitoring required

Behaviour-Informed Disaggregation

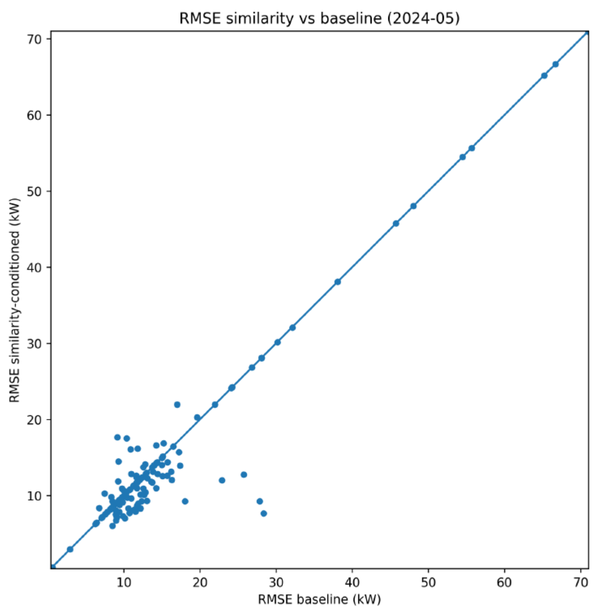


Behavioural Similarity in LV Demand





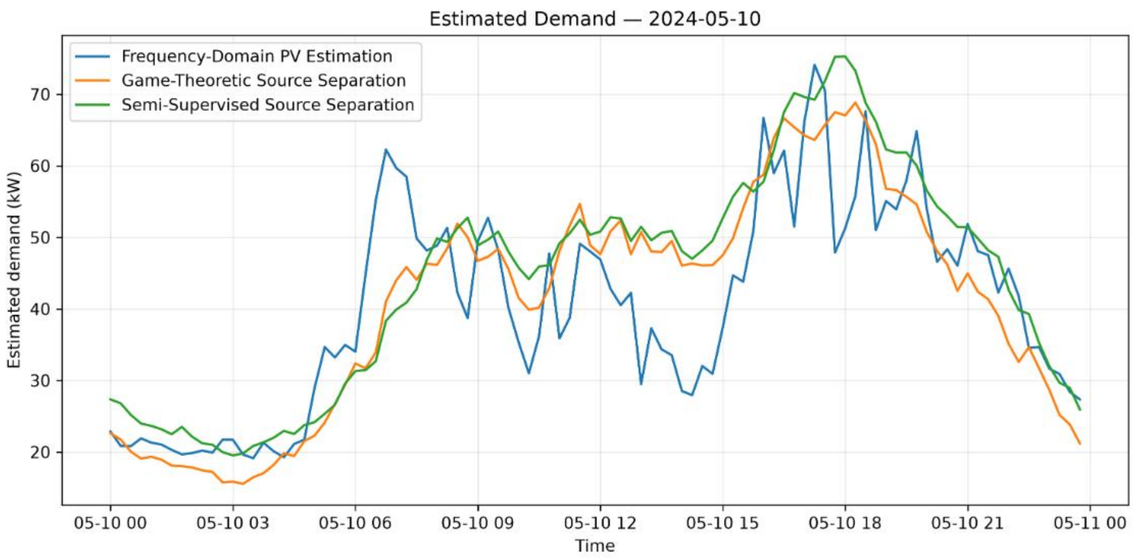
Similarity-Conditioned PV Disaggregation



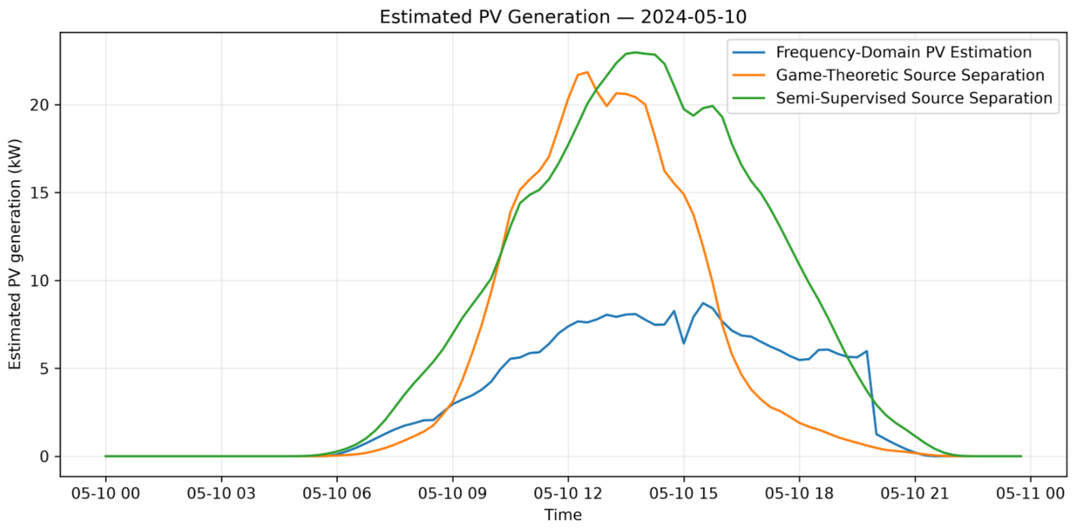
Mean RMSE ↓ 5.1%
14.9 → 14.1kW
Mean MAE ↓ 4.7%
10.9 → 10.5kW

Improvements occur primarily on behaviourally ambiguous days where night-only matching is insufficient

Ongoing Work: Estimated Demand



Ongoing Work: Estimated PV



Key Findings

- Behaviour matters
- Behaviour can improve PV disaggregation
- The problem extends beyond reconstruction accuracy

Next Steps

- Expand evaluation
- Cross-method benchmarking
- Behavioural conditioning
- Validation

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Explore PRISM

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